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MolParetoBO

Surrogate-assisted multi-objective molecular optimization

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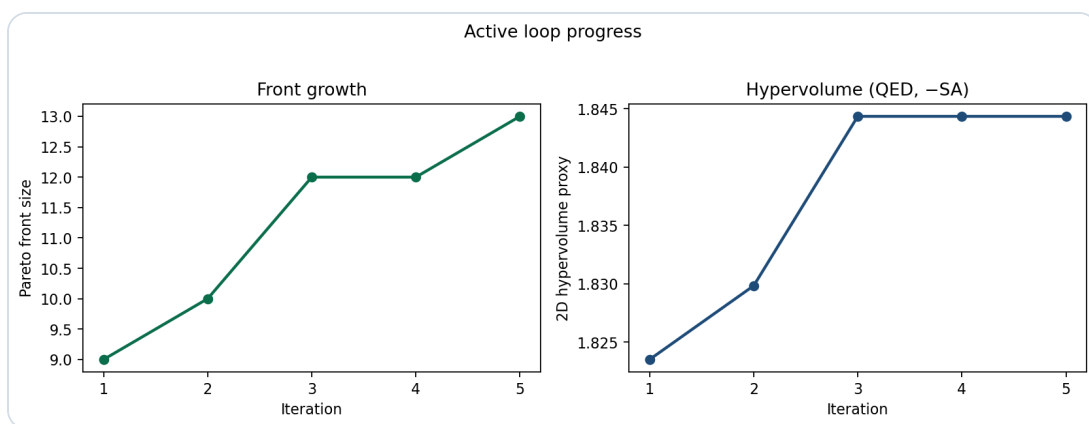
Why multi-objective?

- Conflicting ADMET / drug-likeness goals
- Weighted scores hide trade-offs
- Pareto keeps undominated compromises
- Oracles are expensive → surrogates + acquisition

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Pipeline

1. Mutate (RDKit)
2. Suitability filters
3. Score 5 endpoints
4. Surrogate (RF)
5. Pareto + acquisition loop

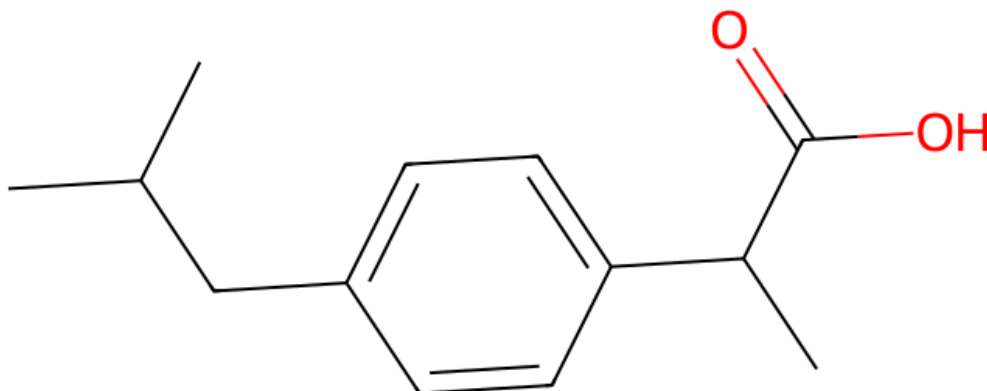


Active loop progress

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Seed & endpoints

Ibuprofen + QED↑ SA↓ logP~2.5 logS↑ Tox↓

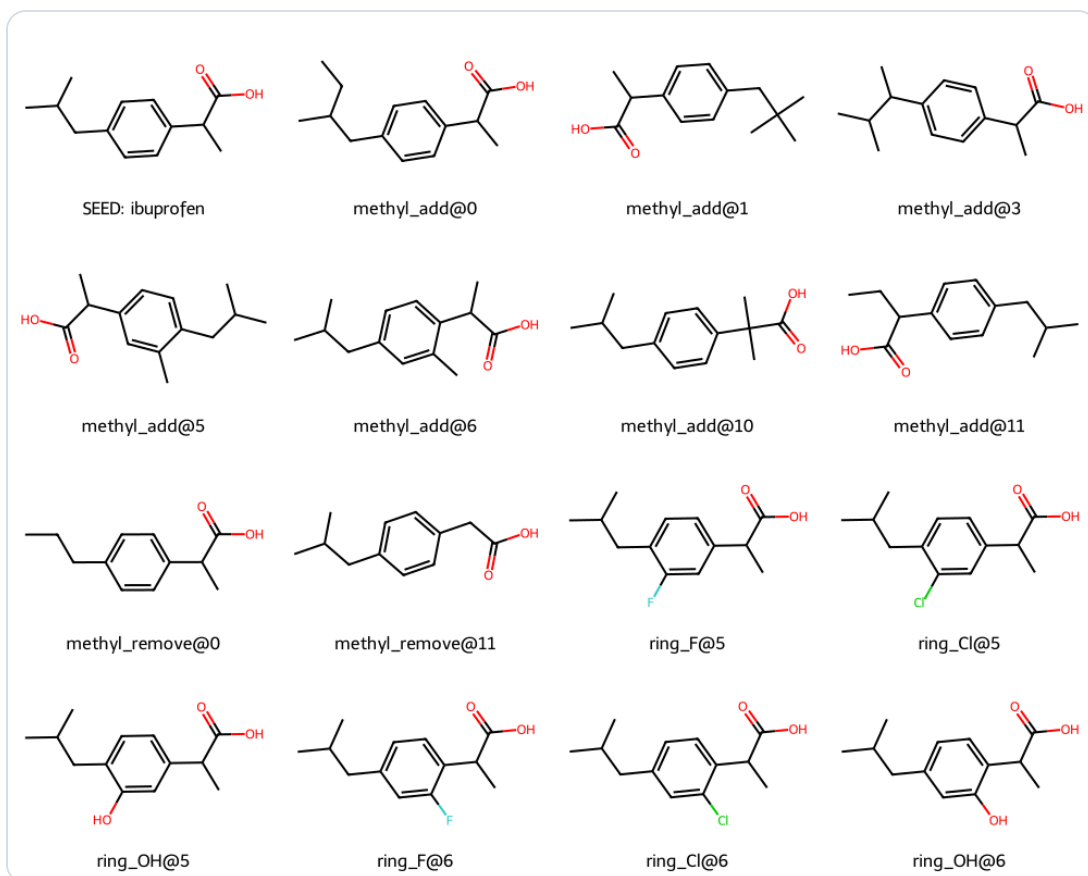


Seed with atom indices

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Mutations & filters

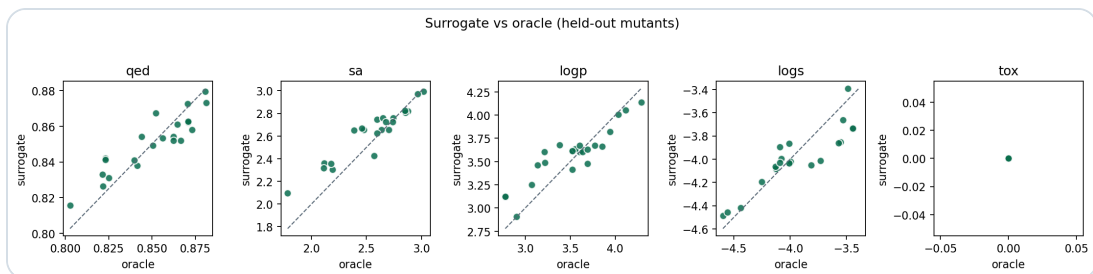
Methyl / ring edits · sanitize · MW · QED gate



Mutation gallery

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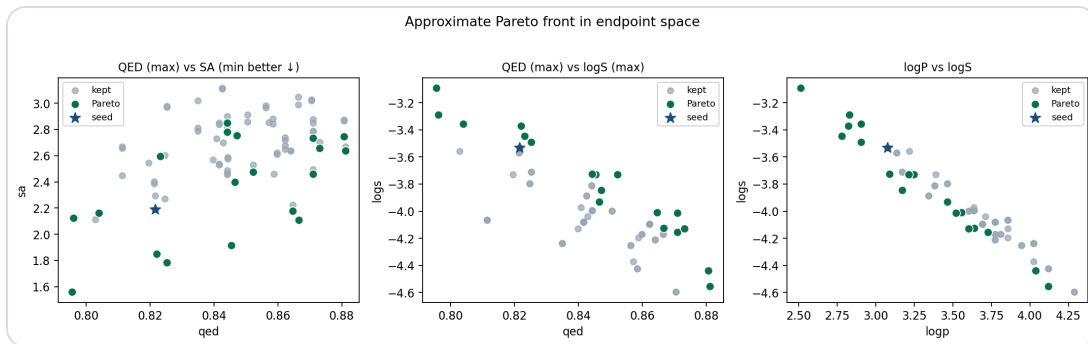
Surrogate



Surrogate vs oracle

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Pareto front



Pareto projections

Acquisition loop

$\alpha \cdot \text{quality} + \beta \cdot \text{uncertainty} + \gamma \cdot \text{diversity}$

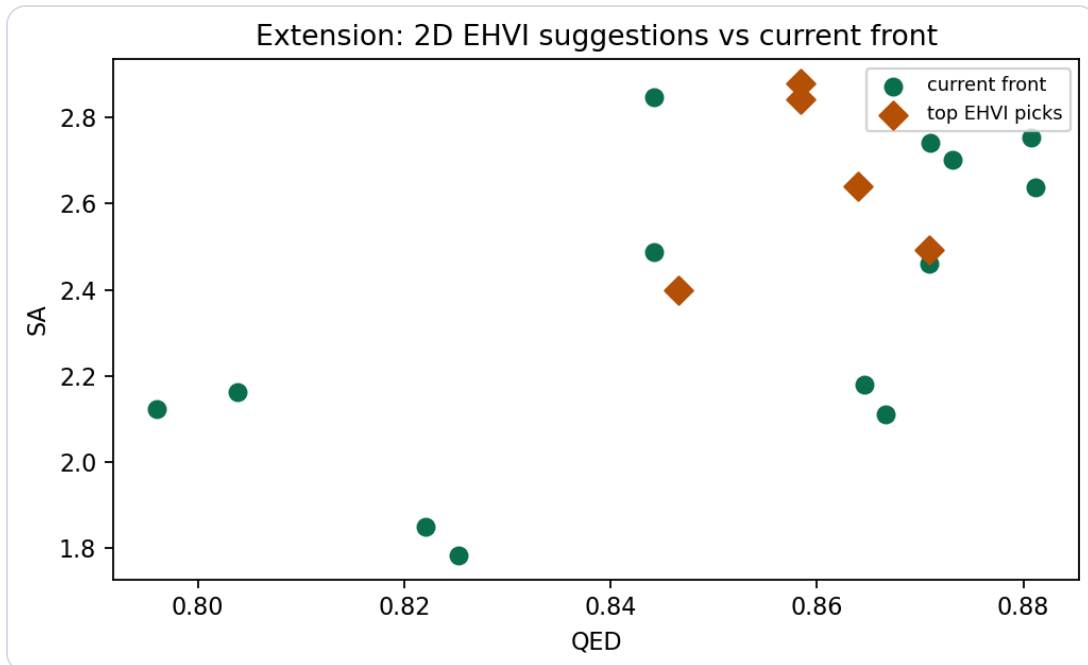
iter	n_labelled	front_size	hv_qed_negSA	batch_mutations	batch_mean_acq
1	13	9	1.823515	gen2lmethyl_remove@0; gen2lmethyl_remove@5; gen2lring_Cl@10; gen2lmethyl_remove@12	0.691293
2	17	10	1.829835	gen2lmethyl_add@1; gen2lring_Cl@7; ring_Cl@5; gen2lring_Cl@7	0.595348
3	21	12	1.844349	gen2lmethyl_add@11; gen2lmethyl_add@4; gen2lmethyl_add@1; gen2lring_Cl@2	0.661508
4	25	12	1.844349	gen2lmethyl_add@0; gen2lmethyl_add@12; gen2lmethyl_add@9; gen2lmethyl_add@9	0.687578
5	29	13	1.844349	gen2lmethyl_remove@0; gen2lmethyl_remove@3; methyl_remove@0; gen2lring_OH@10	0.658671

Discussion

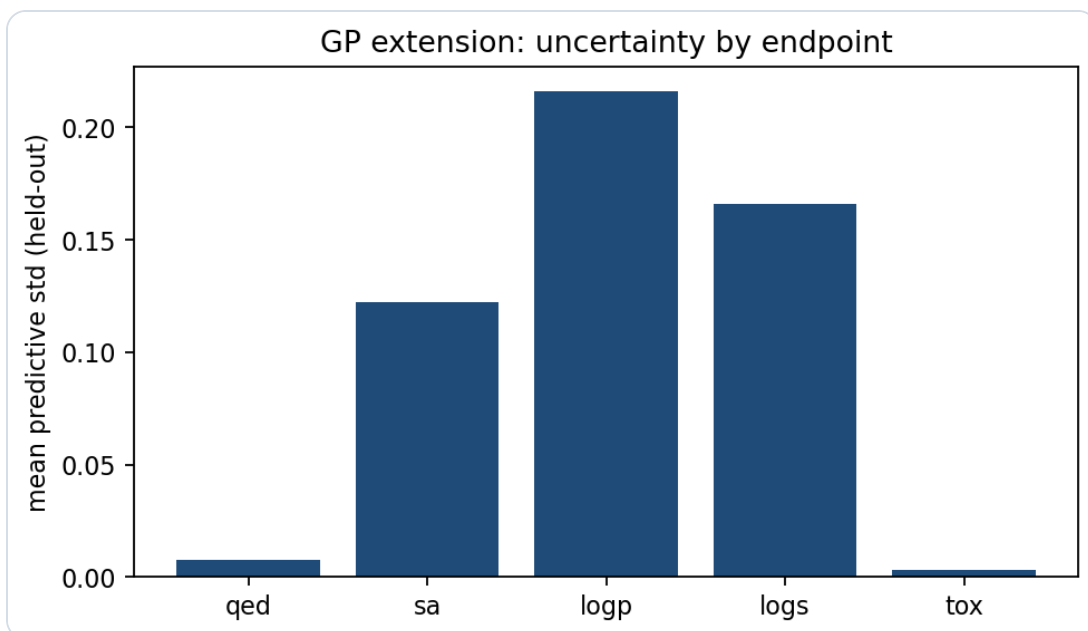
- **Strength:** full reproducible AI workflow + widgets
- **Limit:** proxy oracles, local chemistry, RF uncertainty
- **Meaning:** front = trade-offs, not one best drug

Extensions

- GP + PCA uncertainty
- NSGA-II crowding subset
- 2D EHVI → roadmap to qEHVI/BoTorch



EHVI picks



Tools & widgets

- **RDKit** mutations/descriptors · **sklearn** RF/GP · **pymoo** NSGA-II
- **ipywidgets** Interactive Control Panel (5 checks)
- 1 filters · 2 inspect · 3 surrogate · 4 acquisition · 5 Pareto
- Local Jupyter or Google Colab

Conclusions

- Teaching-scale MolParetoBO delivered
- Colab-ready · report · slides · widgets
- Ready for SoftMining seeds / stronger oracles

Thank you — questions?